

Riyanshi Kedia

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Education

Northeastern University

Master of Science in Information Systems

Boston, MA

Expected May 2027

Relevant Coursework: Data Science, Statistics, Machine Learning, Database Design, Big Data Analytics, Computer Science

Vellore Institute of Technology

Bachelor of Computer Applications (Computer Science) — GPA: 3.73/4.00

Vellore, India

May 2024

Honors: Ranked 6th in graduating class of 200+ students; Academic Merit Scholarship recipient (Top 8 rank, 2023-24)

Technical Skills

Programming: Python, Java, SQL, R, C, C++, PHP, JavaScript, Shell Scripting

Data Science & ML: Scikit-learn, TensorFlow, PyTorch, Prophet, Statistical Modeling, Regression, CNN, LSTM, NLP, RAG, LangChain, Instructor, Pydantic

Data Analysis & Visualization: Power BI, Excel, SPSS, Pandas, NumPy, Matplotlib, Seaborn, Plotly, Streamlit

Database & Cloud: PostgreSQL, MySQL, SQLite, Oracle, SQLAlchemy, Pinecone Vector DB, Google Cloud Platform (GCS, Cloud Run, Cloud Functions, Cloud Build), AWS S3

Development & DevOps: FastAPI, Flask, Docker, Docker Compose, Nginx, Apache Airflow, Git, DVC, MLOps, RESTful APIs, CI/CD, GitHub Actions

Big Data & ETL: Apache Airflow, Web Scraping (Scrapy, BeautifulSoup, Selenium, Playwright, trafilatura), Data Pipelines, ETL Processes

AI & Agent Frameworks: OpenAI GPT-4/GPT-4o, Model Context Protocol (MCP), ReAct Pattern, Multi-Agent Systems, Human-in-the-Loop (HITL)

Work Experience

Operations and Data Analyst

May 2024 – December 2024

Satguru Sai Traders

Nagpur, India

- Automated invoice processing for 100+ monthly PDF documents using Python, reducing manual processing time by 70% and eliminating 95% of data entry errors through custom extraction algorithms and database integration
- Developed predictive Excel model analyzing 5+ years of historical data across 14 coal mining sites, reducing price forecasting time by 60% and improving bid strategy accuracy by 25%, resulting in \$50K+ cost savings
- Analyzed 10,000+ transactional records to identify operational inefficiencies, presenting data-driven recommendations to executive leadership that improved supply chain efficiency by 18%
- Built automated reporting dashboard tracking 15+ key performance indicators (KPIs) for real-time business monitoring, reducing report generation time from 8 hours to 30 minutes
- Managed data quality and validation processes for 50+ data sources, ensuring 99% accuracy in business intelligence reporting

Global Analytics and Reporting Intern

May 2023 – July 2023

D.E. Shaw India Private Limited

Hyderabad, India

- Analyzed 300,000+ candidate applications spanning 5-year period (2018-2022) using Power BI and Excel, identifying hiring trends across 12 job families and 8 geographic regions
- Executed end-to-end ETL processes for 50+ GB of recruiting data, creating 25+ interactive visualizations and dashboards used by 100+ stakeholders across talent acquisition teams
- Designed and implemented 8 automated Power BI dashboards presenting real-time hiring metrics, reducing manual reporting effort by 40 hours per month and enabling data-driven workforce planning
- Collaborated with 6 cross-functional teams to map recruiting workflows and identify 12 process bottlenecks through statistical analysis and process mining

- Delivered actionable insights that reduced average hiring turnaround time (TAT) by 33% (from 45 to 30 days) through optimization of screening and interview scheduling processes, impacting 5,000+ annual hires
- Conducted A/B testing analysis on recruiting strategies across 3 regional offices, identifying best practices that increased offer acceptance rates by 15%

Projects

Project ORBIT — Agentic Private Equity Intelligence Platform | *Python, FastAPI, Streamlit, OpenAI GPT-4o, MCP, Airflow, Docker, GCP, Pinecone* Dec 2024

- Architected production-ready agentic AI system automating private equity intelligence gathering and analysis for 50 Forbes AI companies, processing 5,000+ documents monthly with 95% extraction success rate
- Engineered multi-agent supervisory workflow using Model Context Protocol (MCP) with 8 specialized tools, implementing ReAct reasoning pattern with thought-action-observation loops for transparent decision-making
- Built dual-pipeline architecture processing both structured (Pydantic models + Instructor) and unstructured (RAG + Pinecone vector DB) data, generating standardized 8-section PE dashboards with 94% factual accuracy
- Developed FastAPI MCP server exposing 12+ endpoints for standardized tool/resource/prompt access, implementing JWT authentication, rate limiting, and automatic OpenAPI documentation
- Orchestrated end-to-end ETL workflows using Apache Airflow with 3 production DAGs (initial load, daily updates, agentic generation), managing 50GB+ data with automated task scheduling and dependency management
- Implemented Human-in-the-Loop (HITL) system with Streamlit dashboard for risk detection and approval workflows, pausing execution when concerning signals detected and enabling manual review before proceeding
- Deployed containerized microservices architecture on Google Cloud Platform using Cloud Run for 5 services (MCP Server, Agent Service, API, Frontend, HITL), achieving 99.9% uptime and 200ms p95 latency
- Built CI/CD pipeline with GitHub Actions automating Docker image builds, pytest test suites (80+ tests covering tools, workflows, and API endpoints), and deployment to GCP Cloud Run
- Designed graph-based workflow with conditional branching (Planner → Data Generator → Risk Detector → HITL/Evaluator) achieving 60%+ prediction accuracy through iterative refinement loops
- Implemented comprehensive web scraping system using Playwright, BeautifulSoup, and trafilatura processing 10,000+ pages from Forbes AI 50 company websites with automated retry logic and proxy rotation
- Created structured JSON logging system capturing complete ReAct traces (thought, action, observation) with correlation IDs for debugging and performance monitoring across distributed services
- Integrated OpenAI embeddings (text-embedding-3-small) with Pinecone vector database storing 3072-dimensional embeddings for semantic search with 85%+ relevance accuracy

Project Meridian — Multi-Agent AI Trading Intelligence System | *Python, FastAPI, Next.js, OpenAI GPT-4, GCP* Sept 2024

- Architected scalable multi-agent AI system with 8 specialized LLM agents collaborating through 5-phase pipeline (Data Gathering, Research Debate, Trading Strategy, Risk Assessment, Portfolio Decision) to generate investment recommendations with full reasoning transparency
- Engineered 8 specialized OpenAI GPT-4 agents (MarketAnalyst, FundamentalsAnalyst, InformationAnalyst, Bull/Bear Researchers, RiskManager, PortfolioManager) with custom system prompts and structured JSON outputs using Pydantic schemas for reliable parsing
- Implemented adversarial debate mechanism with iterative 3-round refinement loops between bullish/bearish researchers, reducing single-model bias and hallucination rates by 40% through consensus-building workflows
- Built automated ETL pipelines ingesting real-time market data, financial statements, news articles, and social media sentiment from 4+ APIs, processing 10,000+ news articles and 500+ company financials daily with Apache Airflow orchestration
- Designed data processing workflows computing 15+ technical indicators (Moving Averages, MACD, RSI, Bollinger Bands, Volume indicators) and fundamental metrics (P/E ratios, revenue growth, DCF valuation) stored in PostgreSQL database
- Developed FastAPI REST API with 12+ endpoints handling 100+ queries/hour, implementing JWT authentication, rate limiting (10 requests/min), Server-Sent Events (SSE) for real-time streaming responses, and automatic OpenAPI documentation
- Designed interactive Next.js dashboard with real-time agent visualization, displaying complete reasoning chain, debate transcripts, and confidence scores for each phase with streaming responses and PDF report generation

- Deployed production system on Google Cloud Platform using Cloud Run for containerized microservices, Cloud SQL for PostgreSQL database, and Cloud Storage for document artifacts, achieving 99.9% availability
- Established CI/CD pipeline with GitHub Actions automating Docker image builds, pytest test suites (80+ tests), deployment to GCP Cloud Run, database migrations, and manual production approval gates
- Engineered multi-layer guardrail system with input validation, Pydantic schema enforcement, hallucination detection (cross-referencing LLM metrics against ground truth), and cost monitoring maintaining under \$2 per query through prompt caching (60% cache hit rate)
- Developed comprehensive evaluation framework with rubric-based scoring (clarity, factual accuracy, relevance, confidence calibration), GPT-4-as-judge automated grading, and backtesting over 2020-2024 historical data achieving 60%+ profitable trade accuracy
- Built test suite with 80+ unit tests validating ETL pipelines, API endpoints, and agent workflows, plus integration tests for end-to-end ticker-to-recommendation flow and debate loop convergence
- Implemented intelligent data caching and batch processing strategies reducing API costs by 60% through prompt caching, compression techniques (abbreviations, summaries), and batching multi-ticker analysis

Project Aurelia — Production-Grade RAG System | *Python, FastAPI, Streamlit, Pinecone, OpenAI, Docker, Nginx* Mar 2025

- Engineered production-grade Retrieval-Augmented Generation (RAG) system using FastAPI and Streamlit, implementing 4 search strategies (vector similarity, hybrid keyword+semantic, cross-encoder neural reranking, Reciprocal Rank Fusion) with 85%+ relevance accuracy
- Developed intelligent document processing pipeline with dual parsers (Docling for complex layouts, PyMuPDF for text extraction), 6 chunking strategies (fixed-size, sentence-based, semantic, sliding window, recursive, markdown-aware), and automated metadata extraction
- Implemented advanced document chunking with overlap optimization, metadata enrichment, and OpenAI text-embedding-3-large model (3072 dimensions) processing complex PDFs into high-dimensional embeddings stored in Pinecone vector database
- Integrated OpenAI GPT-4o-mini with cross-encoder neural reranking (top-20 to top-5 refinement), intelligent query routing (FAQ vs document search), PII filtering (SSN, credit card detection), and Wikipedia fallback for general knowledge queries
- Built scalable microservice architecture with comprehensive REST API (8+ endpoints), automatic Swagger documentation, health monitoring endpoints, session management with UUID tracking, and containerized deployment (Docker + Docker Compose)
- Achieved 1.5-4 second end-to-end response times with streaming support, supporting real-time configuration through interactive chat interface with Nginx reverse proxy for production deployment

Investment Report Extractor — Automated Data Pipeline | *Python, Apache Airflow, Scrapy, BeautifulSoup, Selenium* Jan 2025

- Built automated web crawler and data extraction pipeline using Python to systematically scrape investment reports, financial documents, and market data from 15+ financial websites (SEC Edgar, Bloomberg, Reuters, company IR pages), processing 5,000+ documents monthly
- Orchestrated end-to-end ETL workflows using Apache Airflow with 12 DAGs for scheduled data collection (daily SEC filings, hourly news, weekly reports), implementing robust error handling, exponential backoff retry logic, and task dependency management
- Developed multi-threaded web scraping modules leveraging Scrapy for fast concurrent crawling (100+ pages/min), BeautifulSoup for HTML parsing with CSS selectors, and Selenium for JavaScript-rendered content with headless Chrome
- Implemented advanced scraping techniques including rotating proxies, user-agent randomization, request throttling, and session management to avoid IP blocks while extracting structured and unstructured data from diverse report formats (PDF, HTML, XML, XBRL)
- Designed scalable data pipeline architecture with automated task scheduling, dependency management, data validation processes (schema checks, duplicate detection, completeness verification), and PostgreSQL storage with partitioned tables for 50GB+ data
- Achieved 95% extraction success rate across 5,000+ monthly documents, reducing manual data collection time by 90% (from 40 hours to 4 hours), with automated notifications for failed extractions

TradeTransition — End-to-End Trade Forecasting & MLOps System | *Python, Flask, Scikit-learn, Docker, Nginx* Mar 2025

- Analyzed 10+ years of U.S. import data (2012-2022) across 5 key sectors (Electronics, Automotive, Pharmaceuticals, Agriculture, Textiles), processing 2.5 million+ trade records (25GB dataset) to identify sector-specific trends and seasonality patterns
- Built ensemble ML forecasting models combining Random Forest, Gradient Boosting, and XGBoost with 85% prediction accuracy, engineered 25+ features including lagged imports, rolling averages, seasonality indicators, and tariff rate changes
- Deployed production-ready MLOps pipeline with Flask REST API handling 1,000+ requests per hour, implementing 8 endpoints (predict, batch predict, model info, health, metrics), containerized using Docker with multi-stage builds
- Orchestrated services with Docker Compose and served behind Nginx reverse proxy with load balancing, achieving 99.9% uptime and 200ms p95 latency under production load
- Implemented automated model monitoring system tracking 12 performance metrics (accuracy, precision, recall, F1, MAE, RMSE, latency), drift detection using Kolmogorov-Smirnov test, and weekly retraining triggers, achieving 92% production-readiness compliance
- Quantified \$2.3B+ in trade flow disruptions across sectors, analyzing impact of 40%+ tariff increases on import volumes across 25+ trading partner countries, identifying Electronics sector as most vulnerable (35% volume decline)

PDF Parser — Financial Document Extraction System | *Python, OCR, NLP, Docker, DVC, pdfplumber, Tesseract* Jan 2025

- Developed multi-engine PDF parsing framework integrating 5 extraction technologies (pdfplumber for text, Tesseract OCR for scanned documents, Camelot for tables, LayoutLMv3 for complex layouts, Google Document AI) to extract structured data from 10,000+ SEC financial filings
- Achieved 94% extraction accuracy across diverse document formats (10-K, 10-Q, 8-K reports) by engineering intelligent fallback system with quality scoring algorithms and cascading parser selection
- Implemented automated parser selection with confidence scoring (character accuracy, table structure validation), cascading fallback chain, and human-in-the-loop review for low-confidence extractions (under 70%), reducing manual data entry by 85%
- Built Data Version Control (DVC) pipeline with Git integration processing 50+ GB of financial documents, tracking data lineage, versioning parsed outputs, and automating remote storage sync with AWS S3 for collaborative development across 3-person team
- Delivered structured outputs in 4 formats (JSON for APIs, CSV for analytics, Markdown for readability, TXT for simple text) with normalized schemas and automated field validation, reducing financial analysis preparation time from 10 hours to 90 minutes per report
- Integrated NLP modules for entity recognition (company names, financial metrics using spaCy), sentiment analysis of MD&A sections, and key phrase extraction (risk factors, forward-looking statements), enriching parsed outputs with semantic metadata

SmartPortfolio AI — Stock Selection & Return Prediction System | *Python, SQL, PostgreSQL, ML, Prophet, yFinance* Nov 2024

- Built AI-powered investment platform automating stock screening across 500+ S&P 500 equities using machine learning regression models (Random Forest, XGBoost, LSTM) and time series forecasting with Facebook Prophet
- Filtered stocks based on 15+ criteria including P/E ratio, revenue growth, profit margin, debt-to-equity, beta, market cap, dividend yield, analyst ratings, and momentum indicators
- Implemented automated ETL pipeline ingesting daily OHLCV price data for 500+ stocks using yFinance API, scraping S&P 500 constituents from Wikipedia, and fetching fundamentals from Alpha Vantage API
- Stored 5+ years of historical data (10GB+) in PostgreSQL database with optimized indexing on ticker and date columns, implementing data quality checks with duplicate detection and missing value imputation
- Developed return prediction engine improving forecast accuracy by 15% over baseline buy-and-hold strategy through feature engineering (20+ technical indicators including RSI, MACD, Bollinger Bands) and hyperparameter tuning via GridSearchCV
- Backtested models across 3-year period (2020-2023) with 1,000+ simulated trades achieving 55% win rate,

implemented Prophet time series models with custom seasonality and changepoint detection forecasting 30-day returns

- Designed portfolio optimization algorithm using mean-variance theory (Markowitz efficient frontier) and Monte Carlo simulation with 10,000+ iterations, applying constraints (max 20% per stock, min 5 stocks, sector diversification)
- Recommended optimal asset allocations achieving 18% improved Sharpe ratios (1.8 vs 1.2 baseline), calculated risk-adjusted performance metrics including Sortino ratio and max drawdown

DeepLip — Audio-Visual Lip Reading & Speech Recognition | *Python, PyTorch, Computer Vision, CNN, LSTM, OpenCV* Dec 2024

- Developed multimodal deep learning system transcribing spoken words from synchronized lip motion and audio streams using dual-pathway architecture (visual CNN + audio spectrogram CNN) with late fusion strategy
- Achieved 90%+ transcription accuracy on 500+ test samples from LRS2 and LRS3 datasets using sequence-to-sequence decoder with attention mechanism
- Built automated preprocessing pipeline processing 10,000+ video frames per sample, implementing facial landmark detection (dlib 68-point model), lip region extraction with bounding box cropping (48x96 pixel ROI), and frame normalization
- Implemented temporal alignment of audio-visual streams using cross-correlation and data augmentation (random crops, horizontal flips, brightness variations), parallelized with multiprocessing for 40% training speedup
- Designed neural architecture combining 3D CNN (C3D) for spatiotemporal lip motion feature extraction with 512-dimensional embeddings and bidirectional LSTM networks (2 layers, 256 hidden units) for temporal sequence modeling
- Trained on 15 video datasets (LRW, LRS2, LRS3, GRID, TCD-TIMIT) totaling 100+ hours of lip reading data with CTC loss function for sequence alignment, capturing co-articulation effects
- Evaluated model performance using Word Error Rate (WER) and Character Error Rate (CER) metrics, achieving 12% WER improvement over baseline Visual-Only lip reading models (from 45% to 33% WER)
- Conducted ablation studies comparing architecture variants (2D vs 3D CNN, GRU vs LSTM, with/without attention), analyzed error patterns identifying challenges with homophones and occluded speech

Shipping Management System | *HTML, CSS, JavaScript, PHP, MySQL* 2023

- Built full-stack web application managing 500+ monthly shipment orders with real-time tracking, automated invoicing, customer portal, and admin dashboard, reducing manual coordination by 25%

Employee Payroll System | *Java, SQLite* 2023

- Developed payroll management system for 200+ employees with automated salary calculations, tax deductions (federal, state, FICA), overtime tracking, and monthly payslip generation, reducing payroll errors by 40%

Leadership & Extracurricular Activities

Head of Logistics & Operations, VIT Techfest GraVITas

- Led team of 20+ members coordinating logistics for 5,000+ attendee technical festival, managing \$50K+ budget across 30+ events, handling vendor contracts and facility bookings

Vice-Captain, VIT Women's Tennis Team

- 4-time State Championship representative; Managed team of 12 athletes, organized practice schedules, and coordinated inter-university tournaments

Head of Delegate Affairs, Model United Nations

- Coordinated event logistics and registration for 200+ participants across 3-day conference, managed 5-member team, and facilitated 8 committee sessions